JEE Mathematics: Quick Review with Grading and Practice Map - Sequence & Progression (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (133 questions; a question that appears in two of these is counted once): High = 12 or more questions, Medium = 1 to 11 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction of Sequence

Question 01

How do you write T_n when S_n is given (AP)?

Model Answer

$T_1 = S_1$ and $T_n = S_n - S_{n-1}$ for $n \geq 2$. If $S_n = An^2 + Bn$ the sequence is an AP with $T_n = A(2n-1) + B$.

Ex: $S_n = n^2 + 2n \Rightarrow T_1 = 3,\ T_n = 2n + 1$ (also true for $n = 1$, so it is an AP).

Grade - Medium

Practice map, main concept: Sheet: C-I 17 | DPP: 1(1); 4(11) | Question Bank: 35

Also used in: Sheet: J-M 31; C-II 33 | Question Bank: 66

Question 02

Can you write the standard form and the nth term of an AP?

Model Answer

Standard form $a,\ a + d,\ a + 2d, \ldots$ with $d = T_n - T_{n-1}$ (constant).

$$T_n = a + (n-1)d$$

Ex: $3, 7, 11, \ldots$: $T_n = 4n - 1,\ T_{20} = 79$.

Grade - Medium

Practice map, main concept: Sheet: J-M 11, 44; C-II 30 | DPP: 1(5-6) | Question Bank: 77

Also used in: Sheet: J-I 5, 21; J-II 26; J-M 21, 52, 62; J-A 5, 13; C-I 8 | DPP: 1(16) | Question Bank: 31, 39, 54

Question 03

How do you find any term of an AP when any two other terms are given?

Model Answer

From T_p and T_q: $d = \dfrac{T_p - T_q}{p - q}$, then $T_n = T_p + (n-p)d$.

Ex: $T_3 = 7,\ T_7 = 19 \Rightarrow d = 3,\ T_{10} = 28$.

Grade - Medium

Practice map, main concept: Sheet: J-I 4-5; J-II 26; J-M 7, 55; C-I 11 | DPP: 1(11, 16) | Question Bank: 23, 40, 54, 60

Also used in: Sheet: J-I 12; J-M 62 | DPP: 1(3) | Question Bank: 29

Question 04

How do you find the number of terms of an AP when all terms are given?

Model Answer

$$n = \frac{l - a}{d} + 1$$

Ex: $101 + 99 + \cdots + 47$: $n = \dfrac{47 - 101}{-2} + 1 = 28$.

Grade - **Medium**

Practice map, main concept: DPP: 1(4) | Question Bank: 51

Also used in: Sheet: J-I 7-8; J-M 2, 44, 68; J-A 8 | DPP: 1(7, 12); 3(3) | Question Bank: 22, 74

Sum of first n terms of AP

Question 05

Memorize both formulae for S_n of an AP and the derivation of S_n.

Model Answer

$$S_n = \frac{n}{2}\,[\,2a + (n-1)d\,] = \frac{n}{2}\,(a + l)$$

Derivation: write S_n forwards and backwards and add; every pair sums to $2a + (n-1)d$.

Ex: $3 + 7 + 11 + \ldots$ (20 terms): $S_{20} = 10\,[\,6 + 76\,] = 820$.

Grade - **High**

Practice map, main concept: Sheet: J-I 9-10, 12; J-II 3, 33; J-M 21, 29, 32, 48, 58; J-A 2, 5; C-I 12; C-II 31 | DPP: 1(2, 12); 2(2) | Question Bank: 31, 45, 47, 64, 67

Also used in: Sheet: J-I 4, 6, 8; J-II 2, 7, 20; J-M 4, 7, 37, 52, 55, 60, 68, 72, 76; J-A 7, 12; C-II 7, 14, 30, 35 | DPP: 1(3, 7, 10); 2(16); 4(4, 9) | Question Bank: 22, 33, 41-42, 46, 48, 51, 58, 60

Highlights of AP

Question 06

What is the condition when a, b, c are in AP?

Model Answer

$2b = a + c$ (and conversely). b is the AM of a and c.

Ex: $2, x, 8$ in AP $\Rightarrow x = 5$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 3; J-II 28, 34; J-M 1, 51, 66; C-I 18-20; C-II 5, 15 | DPP: 2(22); 3(7); 4(1, 13) | Question Bank: 39, 57

Also used in: Sheet: J-I 11; J-II 4, 32; J-M 4, 12, 47; C-I 4-5, 14 | DPP: 3(12-13) | Question Bank: 27, 34, 58, 73, 78

Question 07

How do you assume 3 / 4 / 5 terms given in AP?

Model Answer

3 terms: $a - d,\ a,\ a + d$. 4 terms: $a - 3d,\ a - d,\ a + d,\ a + 3d$. 5 terms: $a - 2d, \ldots, a + 2d$. The symmetric choice makes the sum simple.

Ex: three numbers in AP, sum 15, product 105: $(5 - d) \cdot 5 \cdot (5 + d) = 105 \Rightarrow d = \pm 2$, numbers $3, 5, 7$.

Grade - **Medium**

Practice map, main concept: Sheet: J-M 53, 64 | Question Bank: 41, 56

Also used in: Sheet: J-I 10; C-I 11

Question 08

How do you identify whether a sequence is an AP when T_n or S_n is given?

Model Answer

T_n linear in n ($T_n = An + B$, $d = A$), i.e. $T_n - T_{n-1}$ is constant. S_n of the form $An^2 + Bn$ with no constant term ($d = 2A$).

Ex: $S_n = 3n^2 + 2n \Rightarrow$ AP with $a = 5,\ d = 6$; but $S_n = 3n^2 + 2n + 1$ is not an AP.

Grade - Medium

Practice map, main concept: Sheet: J-II 2; C-II 7, 29

Also used in: DPP: 2(2); 4(11) | Question Bank: 35

Question 09

When an algebraic operation is done on all terms of an AP, can you decide whether the new sequence is an AP, and use it in questions?

Model Answer

Adding / subtracting / multiplying / dividing every term by the same number gives an AP (new d is multiplied or divided). Adding or subtracting the corresponding terms of two APs gives an AP. Product / quotient of two APs is not an AP in general.

Ex: $3, 7, 11, \ldots$ doubled and increased by 1: $7, 15, 23, \ldots$ (AP, $d = 8$).

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 1 | DPP: 4(13) | Question Bank: 57

Question 10

How do you find the sequence of common terms of two or more APs?

Model Answer

The common terms form an AP: find the first common term; its common difference is $\text{lcm}(d_1, d_2, \ldots)$. Count only terms lying inside every range.

Ex: $17, 21, \ldots, 417$ and $16, 21, 26, \ldots, 466$: common terms $21, 41, \ldots$ ($d = 20$), $21 + 20(n-1) \leq 417 \Rightarrow n = 20$.

Grade - Medium

Practice map, main concept: Sheet: J-M 17, 22, 43, 60; J-A 8, 10 | DPP: 4(7) | Question Bank: 74

Also used in: none

Question 11

How do you find the smallest positive / largest negative term of an AP?

Model Answer

Solve $T_n > 0$ (or < 0) for the least / greatest integer n. The sum S_n is greatest when the last positive term is included.

Ex: $20,\ 19\frac{1}{3},\ 18\frac{2}{3}, \ldots$: $T_n = 20 - \frac{2}{3}(n-1) > 0 \Rightarrow n \leq 30$; $S_{30} = 15\,(40 - \frac{58}{3}) = 310$ is the maximum sum.

Grade - Medium

Practice map, main concept: DPP: 4(9) | Question Bank: 48

Also used in: Sheet: J-II 33; J-A 1 | DPP: 3(4)

Question 12

The ratio of the sums of n terms of two APs is given. How do you find the ratio of their k^{th} terms?

Model Answer

If $\dfrac{S_n}{S'_n} = f(n)$ then $\dfrac{T_k}{T'_k} = f(2k-1)$ (because $S_{2k-1} = (2k-1)\,T_k$).

Ex: $\dfrac{S_n}{S'_n} = \dfrac{2n}{n+1} \Rightarrow \dfrac{T_8}{T'_8} = \dfrac{2 \cdot 15}{16} = \dfrac{15}{8}$.

Grade - Low

Practice map, main concept: Sheet: J-I 6 | DPP: 1(8)

Also used in: none

Question 13

How do you use the sum of terms that are equidistant from the two ends of an AP?

Model Answer

$T_k + T_{n-k+1} = a + l$ (constant); more generally $T_p + T_q = T_r + T_s$ whenever $p + q = r + s$.

Ex: $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225 \Rightarrow 3(a_1 + a_{24}) = 225 \Rightarrow S_{24} = 12 \cdot 75 = 900$.

Grade - Medium

Practice map, main concept: DPP: 1(3)

Also used in: Sheet: J-II 1; J-M 21

Question 14

How do you count or add the numbers lying in a range that are divisible by a or b (or leave a given remainder)?

Model Answer

The multiples of m (or numbers leaving remainder r on division by m) form an AP with $d = m$: count and sum with $n = \frac{l-a}{d} + 1$, S_n. For 'a or b' use inclusion–exclusion: $S(a) + S(b) - S(\text{lcm}(a,b))$; for 'not divisible' subtract from the total.

Ex: integers 1 to 100 divisible by 2 or 5: $2550 + 1050 - 550 = 3050$.

Grade - Medium

Practice map, main concept: Sheet: J-I 7; J-M 2, 5, 16, 19, 37 | DPP: 1(7, 10); 2(1) | Question Bank: 22, 33, 59, 83

Also used in: none

Arithmetic Mean (AM)

Question 15

How do you find the AM of n numbers?

Model Answer

$A = \dfrac{a_1 + a_2 + \cdots + a_n}{n}$, for two numbers $A = \dfrac{a+b}{2}$ (a, A, b in AP).

Ex: AM of $2, 4, 9$ is 5.

Grade - Medium

Practice map, main concept: Sheet: J-II 30

Also used in: Sheet: J-I 20, 22, 25; J-II 9-10; J-A 4 | DPP: 2(19); 3(19)

Question 16
How do you find n AMs between two numbers?

Model Answer

$$d = \frac{b-a}{n+1}, \qquad A_k = a + kd \ (k = 1, \ldots, n)$$

Ex: 4 AMs between 2 and 12: $d = 2$, i.e. $4, 6, 8, 10$.

Grade - Medium

Practice map, main concept: Sheet: J-I 1; J-II 1; C-I 1; C-II 8-9 | DPP: 1(13) | Question Bank: 49
Also used in: Sheet: J-M 34; C-I 21 | Question Bank: 68

Question 17
What is the sum of n AMs between two numbers?

Model Answer

$$A_1 + A_2 + \cdots + A_n = n \cdot \frac{a+b}{2} = nA$$

Ex: 5 AMs between 2 and 12: sum $= 5 \cdot 7 = 35$.

Grade - Low

Practice map, main concept: Sheet: J-I 24 | DPP: 1(9, 14)
Also used in: Sheet: J-II 1

Geometric Progression (GP)

Question 18
Can you write the standard form and the nth term of a GP?

Model Answer
Standard form $a,\ ar,\ ar^2, \ldots$ with $r = \dfrac{T_n}{T_{n-1}}$.

$$T_n = ar^{\,n-1}$$

Ex: $96, 48, 24, \ldots, \frac{3}{16}$: $r = \frac{1}{2}$, $96\,(\frac{1}{2})^{n-1} = \frac{3}{16} \Rightarrow n = 10$.

Grade - High

Practice map, main concept: Sheet: J-I 17-18; J-M 9, 13-14, 33, 46, 49, 69 | DPP: 2(7, 23) | Question Bank: 75, 79-80
Also used in: Sheet: J-I 14-16; J-M 27, 45; C-II 11 | DPP: 2(3); 3(8) | Question Bank: 52, 63, 70, 74

Question 19
What is the condition on r for a GP?

Model Answer
$a \neq 0$, $r \neq 0$ (no term can be zero). For $a > 0$: $r > 1$ increasing, $0 < r < 1$ decreasing, $r < 0$ alternating signs, $r = 1$ constant.

An infinite GP has a sum only when $|r| < 1$.

Grade - Medium

Practice map, main concept: Sheet: C-I 6
Also used in: Sheet: J-II 29; J-M 46 | DPP: 2(9, 23-24)

Sum of n terms of GP

Question 20

Memorize the formulae for S_n of a GP and its derivation.

Model Answer

$$S_n = \frac{a\,(r^n - 1)}{r - 1} = \frac{a\,(1 - r^n)}{1 - r}\ (r \neq 1), \qquad S_n = na\ (r = 1)$$

Derivation: subtract rS_n from S_n; all middle terms cancel, $(1 - r)S_n = a(1 - r^n)$.

Ex: $2 + 6 + 18 + \ldots$ (6 terms) $= \dfrac{2(3^6 - 1)}{2} = 728$.

Grade - Medium

Practice map, main concept: Sheet: J-I 14, 16; J-II 5; J-M 27, 45; J-A 12; C-II 3, 11 | DPP: 2(4, 21); 3(16); 4(12) | Question Bank: 52, 63

Also used in: Sheet: J-II 34; J-M 9, 71; J-A 7, 15 | DPP: 2(16-17, 26); 3(2) | Question Bank: 46, 69, 75

Question 21

Find the sum of an infinite GP and the condition for it to exist.

Model Answer

$$S_\infty = \frac{a}{1 - r} \qquad \text{if } |r| < 1$$

Ex: $1 + \frac{1}{2} + \frac{1}{4} + \cdots = 2$; sum of squares of the terms: $\dfrac{a^2}{1 - r^2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 29; J-II 6, 8, 17, 29; J-M 42, 71; C-I 9; C-II 16, 20 | DPP: 2(8-10, 20, 24-26); 3(18) | Question Bank: 24, 43-44, 88

Also used in: Sheet: J-II 4; J-M 14, 67; C-I 6, 13; C-II 1, 5, 12, 22 | DPP: 2(4); 3(15); 4(5) | Question Bank: 80

Question 22

How do you convert a recurring decimal into a fraction?

Model Answer

Write the repeating part as an infinite GP: $0.\overline{pq} = \dfrac{pq}{100} + \dfrac{pq}{100^2} + \cdots = \dfrac{pq}{99}$.

Ex: $0.\overline{36} = \frac{4}{11}$, $2.\overline{357} = 2 + \frac{357}{999} = \frac{2355}{999}$.

Grade - Low

Practice map, main concept: DPP: 2(6)

Also used in: none

Highlights of GP

Question 23

How do you assume 3 / 4 / 5 terms given in GP?

Model Answer

3 terms: $\frac{a}{r},\ a,\ ar$. 4 terms: $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$ (ratio r^2). 5 terms: $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$. The product of the symmetric terms is easy.

Ex: three numbers in GP, product 27, sum 13: $a = 3,\ \frac{3}{r} + 3r = 10 \Rightarrow r = 3$ or $\frac{1}{3}$; numbers $1, 3, 9$.

Grade - Medium

Practice map, main concept: Sheet: J-I 15; J-M 6; C-II 17 | DPP: 2(11, 18); 4(2) | Question Bank: 27, 32, 69

Also used in: Sheet: J-I 19; J-II 27 | DPP: 2(12)

Question 24
What is the relation between AP and GP?

Model Answer
If a_n is an AP then c^{a_n} $(c > 0)$ is a GP; if $b_n > 0$ is a GP then $\log b_n$ is an AP. A positive sequence that is both an AP and a GP is constant.

Ex: 2nd, 5th, 9th terms of a non-constant AP in GP: $(a + 4d)^2 = (a + d)(a + 8d) \Rightarrow a = 8d$; the terms are $9d, 12d, 16d$, ratio $\frac{4}{3}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 19; J-II 4; J-A 7
Also used in: Sheet: J-II 35

Question 25
What is the condition when a, b, c are in GP?

Model Answer
$b^2 = ac$ (for non-zero terms of the same sign)

If the square of the middle term equals the product of the extremes, the terms are in GP. Ex: $2, x, 18$ in GP $\Rightarrow x = \pm 6$.

Grade - High
Practice map, main concept: Sheet: J-I 21; J-II 35; J-M 4, 12, 52, 54, 62; J-A 4; C-I 4, 7-8 | DPP: 2(3, 12-13); 3(8); 4(10) | Question Bank: 29, 34, 58, 73, 78
Also used in: Sheet: J-I 17; J-II 4, 27, 32, 34; J-M 6, 13, 33, 36, 53, 59, 64, 69; C-II 3, 15, 17 | DPP: 2(11, 18); 3(17); 4(2) | Question Bank: 39, 53, 69, 79, 85

Question 26
How do you convert a GP into an AP?

Model Answer
Take logarithms of the (positive) terms: $\log b_n$ is an AP with common difference $\log r$.

Ex: $x, y, z > 1$ in GP $\Rightarrow \ln x, \ln y, \ln z$ in AP, so $\frac{1}{\ln x}, \frac{1}{\ln y}, \frac{1}{\ln z}$ are in HP.

Grade - Medium
Practice map, main concept: Sheet: J-I 13; J-M 47 | DPP: 3(1)
Also used in: Sheet: J-II 35

Question 27
How do you find the condition when three terms of a GP (or AP) are the sides of a triangle?

Model Answer
Apply the triangle inequality: the sum of the two smaller sides $>$ the largest side. For a, ar, ar^2 with $r > 1$: $1 + r > r^2$.

Ex: $1 < r < \dfrac{1 + \sqrt{5}}{2}$, so $[r] = 1$, $[-r] = -2$ and $3[r] + [-r] = 1$.

Grade - Medium
Practice map, main concept: Sheet: J-M 61
Also used in: Sheet: J-M 59

Sequences convertible to GP

Question 28
Are you able to solve questions based on sequences convertible to GP?

Model Answer
For $a_{n+1} = p\,a_n + q$ add the constant $k = \dfrac{q}{p-1}$: $a_{n+1} + k = p\,(a_n + k)$, so $a_n + k$ is a GP. Also try taking logs of a product-type recurrence.

Ex: $a_1 = 3,\ a_{n+1} = 3a_n + 1 \Rightarrow a_n + \frac{1}{2} = \frac{7}{2}\,3^{n-1} \Rightarrow a_n = \dfrac{7 \cdot 3^{n-1} - 1}{2}$.

Grade - Medium
Practice map, main concept: Sheet: C-II 12
Also used in: Sheet: J-A 9

Geometric Mean (GM)

Question 29
How do you find the GM of n numbers?

Model Answer
$G = (a_1 a_2 \cdots a_n)^{1/n}$, for two numbers $G = \sqrt{ab}$ (a, G, b in GP).
Ex: GM of $2, 4, 8$ is $(64)^{1/3} = 4$.

Grade - Medium
Practice map, main concept: DPP: 2(14-15, 19)
Also used in: Sheet: J-I 22, 25; J-II 9-10, 30; J-M 64 | DPP: 3(6, 19)

Question 30
How do you find n GMs between two numbers?

Model Answer

$$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}, \qquad G_k = a\,r^k$$

Ex: 3 GMs between 3 and 48: $r = 2$, i.e. $6, 12, 24$.

Grade - Medium
Practice map, main concept: Sheet: J-I 20; J-M 34 | Question Bank: 68
Also used in: Sheet: C-I 21 | Question Bank: 49

Question 31
What is the product of n GMs between two numbers?

Model Answer

$$G_1 G_2 \cdots G_n = (ab)^{n/2} = G^n$$

Ex: $6 \cdot 12 \cdot 24 = 1728 = (3 \cdot 48)^{3/2}$. Also $G_k G_{n+1-k} = ab$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 24; C-I 1

Concept of AM-GM

Question 32

State AM $\geq$ GM for any n positive numbers.

Model Answer

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \;\geq\; (a_1 a_2 \cdots a_n)^{1/n} \qquad (a_i > 0)$$

Ex: $x > 0 \Rightarrow x + \dfrac{1}{x} \geq 2$; for two numbers $\dfrac{a+b}{2} \geq \sqrt{ab}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 25

Also used in: Sheet: J-I 23, 26; J-II 11; J-M 51, 66; J-A 7; C-I 19, 22; C-II 21 | Question Bank: 32

Question 33

When does AM = GM hold?

Model Answer

Only when all the numbers are equal: $a_1 = a_2 = \cdots = a_n$. This gives the point where the minimum / maximum occurs.

Ex: $x + \dfrac{4}{x} \geq 4$ with equality at $x = 2$.

Grade - Low

Practice map, main concept: Sheet: C-I 22

Also used in: Sheet: J-II 12

Question 34

When do we think of applying the AM-GM concept in questions?

Model Answer

When the variables are positive and a sum / product is to be minimised / maximised, or an inequality (with $>$, $\geq$, 'prove that') between sums and products is to be shown. Split terms (with weights) so that the product becomes a constant.

Ex: $a, b, c > 0 \Rightarrow (a+b)(b+c)(c+a) \geq 8abc$, by AM-GM on each bracket.

Grade - Medium

Practice map, main concept: Sheet: J-II 11; C-II 21

Also used in: Sheet: J-M 30; J-A 6; C-I 19, 22; C-II 26-27 | Question Bank: 25, 55, 65

Question 35

What should you keep in mind while minimising a sum using AM-GM?

Model Answer

The product of the chosen terms must be a constant (no variable left) and the terms must be able to become equal.

Ex: $x^3y^2 = 2^{15}$: $3x + 2y = x + x + x + y + y \geq 5\,(x^3y^2)^{1/5} = 40$, equality at $x = y = 8$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23, 26; J-II 12; J-A 6, 11 | Question Bank: 55

Also used in: none

Question 36
What should you keep in mind while maximising a product using AM-GM?

Model Answer
The sum of the chosen terms must be a constant; divide each variable by its power so that all parts can become equal.

Ex: $a+b+c+d=11$: $\frac{a}{5}\times 5,\ \frac{b}{3}\times 3,\ \frac{c}{2}\times 2,\ d$ give $a^5b^3c^2d \le 5^5\,3^3\,2^2 = 337500$.

Grade - Medium

Practice map, main concept: Sheet: J-M 30; C-II 26-28 | Question Bank: 25, 65
Also used in: Sheet: J-A 11

Harmonic Progression (HP)

Question 37
What is the basic definition of an HP?

Model Answer
A sequence is an HP if the reciprocals of its terms are in AP (no term may be zero): $T_n = \dfrac{1}{a+(n-1)d}$.

Ex: $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$ is an HP (reciprocals $2, 4, 6, \dots$).

Grade - Low

Practice map, main concept: Sheet: J-II 32; C-I 5
Also used in: Sheet: C-I 20, 24; C-II 15

Question 38
How do you solve questions on HP by converting them into the corresponding AP?

Model Answer
Take reciprocals of all the given terms, solve the AP, then take the reciprocal of the answer. a, b, c in HP $\Leftrightarrow b = \dfrac{2ac}{a+c}$.

Ex: p^{th} term q, q^{th} term p: $\frac{1}{q} = a+(p-1)d,\ \frac{1}{p} = a+(q-1)d \Rightarrow a = d = \frac{1}{pq}$, so $T_{pq} = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 28; J-II 13-14; J-M 36; J-A 1; C-I 14, 24 | DPP: 3(3-5, 9-13) | Question Bank: 85
Also used in: Sheet: J-I 13, 29; C-I 8; C-II 10 | DPP: 3(1)

Harmonic Mean (HM)

Question 39
How do you find the HM of n numbers?

Model Answer
$H = \dfrac{n}{\frac{1}{a_1}+\frac{1}{a_2}+\cdots+\frac{1}{a_n}}$, for two numbers $H = \dfrac{2ab}{a+b}$.

Ex: HM of 3 and 6 is $\dfrac{2\cdot 18}{9} = 4$.

Grade - Low

Practice map, main concept: Sheet: J-I 22, 27; C-I 25 | DPP: 3(14)
Also used in: Sheet: J-II 9; C-I 1; C-II 6, 35 | DPP: 3(6)

Question 40
How do you find n HMs between two numbers?

Model Answer

$$\frac{1}{H_k} = \frac{1}{a} + k\,d, \qquad d = \frac{\frac{1}{b} - \frac{1}{a}}{n+1}$$

Ex: 2 HMs between 3 and 12: $d = -\frac{1}{12}$, $\frac{1}{H_1} = \frac{1}{4}$, $\frac{1}{H_2} = \frac{1}{6}$, so $3, 4, 6, 12$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-I 21; C-II 8-9

Question 41
What is the sum of the reciprocals of n HMs between two numbers?

Model Answer

$$\sum_{k=1}^{n} \frac{1}{H_k} = n \cdot \frac{\frac{1}{a} + \frac{1}{b}}{2}$$

Ex: $\frac{1}{4} + \frac{1}{6} = \frac{5}{12} = 2 \cdot \frac{\frac{1}{3} + \frac{1}{12}}{2}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Relation between AM, GM and HM

Question 42
What is the relation between RMS, AM, GM and HM for n positive numbers?

Model Answer

$$\text{RMS} \geq \text{AM} \geq \text{GM} \geq \text{HM}, \qquad \text{RMS} = \sqrt{\frac{a_1^2 + \cdots + a_n^2}{n}}$$

Equality holds only when all the numbers are equal. Ex: $1, 4$: $\text{AM} = 2.5,\ \text{GM} = 2,\ \text{HM} = 1.6$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 23; J-II 10, 23-25; C-II 6

Question 43
Relation $G^2 = AH$ for two numbers and the condition when it is applicable for n positive numbers.

Model Answer
For two positive numbers $G^2 = AH$ (A, G, H are in GP). For $n > 2$ numbers it holds only when the numbers are in GP.
Ex: $A = 6$ and $G^2 + 3H = 48$: $G^2 = 6H \Rightarrow H = \frac{16}{3}$, $G^2 = 32$; $a + b = 12$, $ab = 32$, numbers $4, 8$.

Grade - Low
Practice map, main concept: Sheet: J-II 9-10, 23-25; C-I 21 | DPP: 3(6, 19)
Also used in: none

Arithmetico-Geometric Progression (AGP)

Question 44

How do you identify whether a sequence is an AGP or an extended AGP from its general term or the sequence?

Model Answer

AGP: each term is (a term of an AP) $\times$ (a term of a GP), e.g. $1, 2x, 3x^2, \ldots$ or $\frac{1}{2}, \frac{3}{4}, \frac{5}{8}, \ldots$ (numerators in AP, denominators in GP). Extended AGP: the AP part is a polynomial in n, e.g. $\dfrac{n^2+2n+3}{2^n}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 45

Are you able to write the general term of an AGP?

Model Answer

$$T_n = [\,a + (n-1)d\,]\, r^{\,n-1}$$

Ex: $1 + 2x + 3x^2 + \ldots$: $T_n = n\,x^{n-1}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 38 | Question Bank: 86

Also used in: Sheet: J-I 30-31; J-M 39, 56; C-II 2 | DPP: 4(15)

Question 46

How do you find the sum of n terms of an AGP or extended AGP?

Model Answer

Write S_n, multiply by r and subtract: $(1-r)S_n = a + d\,(r + r^2 + \cdots + r^{n-1}) - [a + (n-1)d]\, r^n$. For an extended AGP repeat the process until the AP part is constant.

Ex: $1 + 2x + 3x^2 + \cdots + nx^{n-1} = \dfrac{1 - (n+1)x^n + nx^{n+1}}{(1-x)^2}$; at $n = 3, x = 2$: $1 + 4 + 12 = 17$.

Grade - Medium

Practice map, main concept: Sheet: J-I 31; J-II 16; J-M 35, 39, 74

Also used in: Sheet: J-M 38, 73 | Question Bank: 86

Question 47

How do you find the sum of infinite terms of an AGP or extended AGP?

Model Answer

$$S_\infty = \frac{a}{1-r} + \frac{d\,r}{(1-r)^2} \qquad (|r| < 1)$$

Ex: $1 - \frac{3}{2} + \frac{5}{4} - \frac{7}{8} + \ldots$: $a = 1,\ d = 2,\ r = -\frac{1}{2} \Rightarrow S = \frac{2}{3} - \frac{4}{9} = \frac{2}{9}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 30; J-II 15; J-M 40, 56; C-II 2, 22, 35 | DPP: 4(6, 15) | Question Bank: 87

Also used in: none

Miscellaneous Sequence Type 1

Question 48

Memorize the results for the sum of the first n natural numbers, their squares and their cubes, and their use in finding S_n when T_n is given.

Model Answer

$$\sum n = \frac{n(n+1)}{2}, \qquad \sum n^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum n^3 = \left[\frac{n(n+1)}{2}\right]^2, \qquad \sum c = nc$$

If T_n is a polynomial in n, split it and add each part. Ex: $T_r = r(r+1) \Rightarrow S_n = \dfrac{n(n+1)(n+2)}{3}$.

Grade - High

Practice map, main concept: Sheet: J-I 36; J-II 31; J-M 25, 63; C-I 3 | DPP: 2(5); 4(17) | Question Bank: 28

Also used in: Sheet: J-I 2, 32, 34-35, 37, 39; J-II 2, 15, 17, 20, 22; J-M 10, 20, 23-24, 26, 28-29, 32, 41, 57-58; J-A 3, 14; C-I 10, 17; C-II 14, 31 | DPP: 4(14) | Question Bank: 26, 30, 36-38, 50, 61-62, 64, 67, 72, 76, 84

Question 49

Breaking the general term while finding S_n by taking similar sequences together.

Model Answer

Split into AP, GP and power-sum parts, or take odd-placed and even-placed terms separately, or group alternating signs.

Ex: $1^2 - 2^2 + 3^2 - \cdots + (-1)^{n-1}n^2 = (-1)^{n-1}\dfrac{n(n+1)}{2}$; for $n = 2003$ this is 2007006.

Grade - Medium

Practice map, main concept: Sheet: J-I 2, 32-33, 37; J-M 20, 24, 41, 72; J-A 3; C-I 2, 10 | DPP: 2(16-17); 3(2); 4(4-5) | Question Bank: 42, 46, 61, 84

Also used in: Sheet: J-II 31; J-M 42, 63 | Question Bank: 44, 88

Question 50

Writing the general term by identifying the pattern of AP / GP / AGP.

Model Answer

Look at the differences or ratios of the terms (or of the numerators and denominators separately), then write T_n and sum it.

Ex: $3 + 33 + 333 + \ldots$: $T_n = \dfrac{1}{3}(10^n - 1)$, so $S_n = \dfrac{10^{n+1} - 10 - 9n}{27}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 34-35, 39; J-M 10; J-A 15 | DPP: 4(14) | Question Bank: 26, 30, 37-38, 50, 76

Also used in: none

Question 51

How do you find the sum of the products of numbers taken two at a time?

Model Answer

$$\sum_{i<j} a_i a_j = \frac{(\sum a_i)^2 - \sum a_i^2}{2}$$

Ex: for $1, 2, \ldots, 20$: $\dfrac{210^2 - 2870}{2} = 20615$. The coefficient of x^8 in $(x-1)(x-2)\cdots(x-10)$ is $\dfrac{55^2 - 385}{2} = 1320$.

Grade - Low

Practice map, main concept: Sheet: J-I 38; J-II 22
Also used in: none

Miscellaneous Sequence Type 2

Question 52

How do you identify whether a sequence is of type 2?

Model Answer

The differences of consecutive terms $u_2 - u_1,\ u_3 - u_2, \ldots$ form an AP (or a GP).

Ex: $3, 7, 13, 21, \ldots$ has differences $4, 6, 8, \ldots$ (AP).

Grade - Medium

Practice map, main concept: Sheet: J-A 13; C-II 32-33
Also used in: Sheet: J-II 18; J-M 15, 23, 26, 28, 40, 57; J-A 14 | Question Bank: 62, 72, 81, 87

Question 53

How do you find the nth term of a type 2 sequence and then S_n with the help of type 1?

Model Answer

Write S twice, shifted by one place, and subtract: $T_n = u_1 + (u_2 - u_1) + \cdots + (u_n - u_{n-1})$; sum this AP / GP, then add T_n using type 1.

Ex: $3 + 7 + 13 + 21 + \ldots$: $T_n = n^2 + n + 1$, $S_n = \dfrac{n\,(n^2 + 3n + 5)}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 18; J-M 15, 23, 26, 28, 57; J-A 14 | Question Bank: 62, 72, 81
Also used in: Sheet: J-M 40 | Question Bank: 87

Miscellaneous Sequence Type 3

Question 54

Explain the basic concept of the method of difference.

Model Answer

Write $T_r = f(r) - f(r-1)$; then $T_1 = f(1) - f(0), \ldots, T_n = f(n) - f(n-1)$ and the sum telescopes to $S_n = f(n) - f(0)$.

Ex: $\dfrac{1}{r(r+1)} = \dfrac{1}{r} - \dfrac{1}{r+1} \Rightarrow S_n = \dfrac{n}{n+1}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 19
Also used in: Sheet: J-II 21; J-M 3, 18, 65, 70, 77; C-I 16; C-II 18, 23 | DPP: 4(8, 16) | Question Bank: 82

Question 55

What do you do after splitting the general term as a difference of two terms in the method of difference?

Model Answer

Write the first few and last few terms one below the other, cancel the middle, and keep only the surviving first and last parts (check the signs). For the infinite sum let $n \to \infty$.

Ex: $\sum \dfrac{1}{r(r+1)(r+2)} = \dfrac{1}{4} - \dfrac{1}{2(n+1)(n+2)} \ \to\ \dfrac{1}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 65, 77; C-II 18
Also used in: Sheet: J-M 8, 50; C-II 20, 23 | DPP: 4(3)

Question 56
How do you identify whether a sequence is of type 3?

Model Answer
Every term has a constant numerator and a denominator that is a product of factors in AP, and the first factor of the terms is also in AP.
Ex: $\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\dots$ and $\frac{1}{1\cdot 3\cdot 5}+\frac{1}{3\cdot 5\cdot 7}+\dots$

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 36

Question 57
What should you keep in mind while splitting the general term as a difference of two terms for type 3?

Model Answer
Remove one factor from each end: for r factors with step d,

$$\frac{1}{a_n a_{n+1}\cdots a_{n+r-1}}=\frac{1}{(r-1)d}\left[\frac{1}{a_n\cdots a_{n+r-2}}-\frac{1}{a_{n+1}\cdots a_{n+r-1}}\right]$$

Ex: $\frac{1}{n(n+1)(n+2)}=\frac{1}{2}\left[\frac{1}{n(n+1)}-\frac{1}{(n+1)(n+2)}\right]$.

Grade - Medium

Practice map, main concept: Sheet: J-II 21; J-M 3, 18, 70; C-II 23 | DPP: 4(8) | Question Bank: 82

Also used in: Sheet: J-II 36; J-M 65

Question 58
What do you do if a factor in the denominator is "missing" in type 3?

Model Answer
Treat the factors as an AP with their own step d (the in-between factor is not needed); use the same formula with that d. If the numerator is not constant, express it as the difference of the extreme factors.
Ex: $\frac{1}{r(r+2)(r+4)}=\frac{1}{4}\left[\frac{1}{r(r+2)}-\frac{1}{(r+2)(r+4)}\right]$, so $\sum_{r\geq 1}=\frac{1}{4}\left(\frac{1}{3}+\frac{1}{8}\right)=\frac{11}{96}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Miscellaneous Sequence Type 4

Question 59
How do you identify whether a sequence is of type 4?

Model Answer
The general term is a product of r factors in AP in the numerator (a polynomial like $n(n+1)(n+2)$).
Ex: $1\cdot 2\cdot 3\cdot 4+2\cdot 3\cdot 4\cdot 5+\dots$ and $1\cdot 3\cdot 5+3\cdot 5\cdot 7+\dots$

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 31 | Question Bank: 66

Question 60

What should you keep in mind while splitting the general term as a difference of two terms for type 4?

Model Answer

Multiply and divide by (step) $\times$ (number of factors $+1$) and add one factor at the end (next factor) minus one factor before the start:

$$T_n = \frac{1}{5}\left[\,n(n{+}1)(n{+}2)(n{+}3)(n{+}4) - (n{-}1)n(n{+}1)(n{+}2)(n{+}3)\,\right] \;\Rightarrow\; S_n = \frac{n(n+1)(n+2)(n+3)(n+4)}{5}$$

Ex: $n=1$: $24 = \dfrac{120}{5}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 31 | Question Bank: 66

Also used in: none

Miscellaneous

Question 61

How do you solve questions similar to the method of difference expressed in continued-product form?

Model Answer

Split T_r into the difference of two consecutive continued products, $T_r = \dfrac{1}{2}\left[\,P_{r-1} - P_r\,\right]$.

Ex: $\dfrac{1}{1\cdot 3} + \dfrac{2}{1\cdot 3\cdot 5} + \dfrac{3}{1\cdot 3\cdot 5\cdot 7} + \ldots$: $T_r = \dfrac{1}{2}\left[\dfrac{1}{1\cdot 3\cdots(2r-1)} - \dfrac{1}{1\cdot 3\cdots(2r+1)}\right]$, $S_\infty = \dfrac{1}{2}$.

Grade - Low

Practice map, main concept: Sheet: J-II 36; C-II 19

Also used in: Sheet: J-II 21

Question 62

How do you find the sum of an infinite sequence when a recurrence relation between the terms is given?

Model Answer

Let $p = \sum a_n x^n$. Multiply the recurrence by the right power of x, sum over n, write every sum in terms of p and solve the linear equation for p.

Ex: $a_1 = a_2 = 1,\ a_{n+2} = 2a_{n+1} + a_n$, $p = \sum \dfrac{a_n}{8^n}$: $64\left(p - \frac{1}{8} - \frac{1}{64}\right) = 16\left(p - \frac{1}{8}\right) + p \Rightarrow p = \dfrac{7}{47}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 73; J-A 9; C-I 23; C-II 24-25

Also used in: none

Question 63

How do you solve a cubic when its roots are in AP / GP / HP?

Model Answer

Take the roots as $\alpha - d, \alpha, \alpha + d$ (AP), $\frac{\alpha}{r}, \alpha, \alpha r$ (GP) or reciprocals of an AP (HP). The middle root follows from Vieta: AP $\alpha = -\frac{b}{3a}$; GP $\alpha^3 = -\frac{d}{a}$; HP: put $x = \frac{1}{y}$ and use the AP rule.

Ex: $2x^3 - 19x^2 + 57x - 54 = 0$ with roots in GP: $\alpha^3 = 27 \Rightarrow \alpha = 3$, roots $2, 3, \frac{9}{2}$.

Grade - Low

Practice map, main concept: Sheet: J-II 27; C-II 1, 6, 10

Also used in: none

Question 64

Practising miscellaneous questions based on the method of difference.

Model Answer

Useful splits: $\dfrac{1}{\sqrt{k}+\sqrt{k+1}} = \sqrt{k+1}-\sqrt{k}$; $\dfrac{r}{(r+1)!} = \dfrac{1}{r!} - \dfrac{1}{(r+1)!}$; $\dfrac{r}{r^4+r^2+1} = \dfrac{1}{2}\left[\dfrac{1}{r^2-r+1} - \dfrac{1}{r^2+r+1}\right]$.

Ex: $\sum_{r=1}^{n} \dfrac{r}{(r+1)!} = 1 - \dfrac{1}{(n+1)!}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 8, 50; C-I 16 | DPP: 4(3, 16)

Also used in: Sheet: J-M 77

Question 65

How do you handle terms written in groups (the nth group, rows of numbers)?

Model Answer

Find the number of terms in the first $n-1$ groups (usually an AP of group sizes), then the first term of the nth group, then its sum with S_n.

Ex: $(1), (2,3,4), (5,\ldots,9), \ldots$: sizes $2n-1$, first term $(n-1)^2+1$, sum $(n-1)^3+n^3$.

Grade - Medium

Practice map, main concept: Sheet: J-II 20; J-M 75-76; C-II 14, 34

Also used in: none

Question 66

How do you solve real-life and geometrical problems that lead to an AP or a GP?

Model Answer

Identify what repeats (distance, count, side, area, perimeter), find the first term and the constant difference / ratio, then use T_n, S_n or S_∞.

Ex: a ball dropped from 100 m rebounds $\frac{4}{5}$ of the height each time: total distance $100 + \dfrac{2\cdot 100\cdot \frac{4}{5}}{1-\frac{4}{5}} =$ 900 m.

Grade - Medium

Practice map, main concept: Sheet: J-I 8; J-II 7; J-M 67-68; C-I 13 | DPP: 3(15) | Question Bank: 36, 70

Also used in: none

Question 67

How do you use "a sum of squares equals zero" to get relations among a, b, c?

Model Answer

Rewrite the given relation as $(\ldots)^2+(\ldots)^2=0$ (real numbers), so each bracket is zero; this often gives $b^2=ac$ or $2b=a+c$.

Ex: $(a^2+b^2)x^2-2b(a+c)x+(b^2+c^2)=0$ with real x: $(ax-b)^2+(bx-c)^2=0 \Rightarrow b^2=ac$, so a, b, c are in GP.

Grade - Medium

Practice map, main concept: Sheet: J-I 11; J-M 59 | DPP: 3(17) | Question Bank: 53

Also used in: none